

1. When solving an equation, the goal is to isolate the variable.

Two students, James and Joe, were solving the equation $6 - 5(11 - 2x) = 17x$. Their work is shown below for questions 2 – 5.

James' Work	Joe's Work
$6 - 5(11 - 2x) = 17x$	$6 - 5(11 - 2x) = 17x$
$\frac{6 - 5(11 - 2x)}{6} = \frac{17x}{6}$	$6 - 55 + 10x = 17x$
$-5(11 - 2x) = \frac{17x}{6}$	$-49 + 10x = 17x$
	$\underline{-10x \quad -10x}$
$-55 + 10x = \frac{17x}{6}$	$\frac{-49}{7} = \frac{7x}{7}$
$\underline{-10x \quad -10x}$	$-7 = x$
$-55 = -\frac{43x}{6}$	
$\underline{\times 6 \quad \times 6}$	
$-330 = -43x$	
$\frac{330}{43} = x$	

2. Describe the different first steps the students used in solving the equation.
James divided everything by 6. Meanwhile, Joe distributed -5 over the subtraction group, $(11 - 2x)$.
3. Which student made an error?
James made an error.

4. Write a note to the student who made an error explaining the mistake they made.

Answers may vary.

Subtract 6 instead and then divide by -5 .

5. Is 8 a solution to the equation $3(x - 2) + 5 = 23$? Justify your answer mathematically.

Yes.

$$3(8 - 2) + 5 = 23$$

$$3(6) + 5 = 23$$

$$18 + 5 = 23$$

$$23 = 23 \checkmark$$

For questions 6 – 10, solve each equation and verify the solution using substitution.

6. $-5(2r - 0.3) + 0.5(4r + 3) = -64$

$$-10r + 1.5 + 2r + 1.5 = -64$$

$$-8r + 3 = -64$$

$$-8r = -67$$

$$r = 8.375 \text{ or } 8\frac{3}{8}$$

$$-5(2(8.375) - 0.3) + 0.5(4(8.375) + 3) = -64$$

$$-5(16.45) + 0.5(36.5) = -64$$

$$-82.25 + 18.25 = -64$$

$$-64 = -64 \checkmark$$

7. $\frac{3m+5}{3} = 9m + 2$

$$3\left(\frac{3m+5}{3} = 9m + 2\right)$$

$$3m + 5 = 27m + 6$$

$$-1 = 24m$$

$$-\frac{1}{24} = m$$

$$\frac{3\left(-\frac{1}{24}\right)+5}{3} = 9\left(-\frac{1}{24}\right) + 2$$

$$-\frac{1}{24} + \frac{5}{3} = -\frac{9}{24} + 2$$

$$-\frac{1}{24} + \frac{40}{24} = -\frac{9}{24} + \frac{48}{24}$$

$$\frac{39}{24} = \frac{39}{24} \checkmark$$

8. $\frac{2}{3} + 13k - 1 = 3k - \frac{5}{3}$

$$13k - \frac{1}{3} = 3k - \frac{5}{3}$$

$$13k = 3k - \frac{4}{3}$$

$$10k = -\frac{4}{3}$$

$$k = -\frac{2}{15}$$

$$\frac{2}{3} + 13\left(-\frac{2}{15}\right) - 1 = 3\left(-\frac{2}{15}\right) - \frac{5}{3}$$

$$-\frac{26}{15} - \frac{1}{3} = -\frac{6}{15} - \frac{5}{3}$$

$$-\frac{26}{15} - \frac{5}{15} = -\frac{6}{15} - \frac{25}{15}$$

$$-\frac{31}{15} = -\frac{31}{15} \checkmark$$

9. $4.1 - 7.2z + 3z = 5.2 + z$

$$-7.2z + 3z - z = 5.2 - 4.1$$

$$-5.2z = 1.1$$

$$z = -\frac{11}{52}$$

$$4.1 - 7.2\left(-\frac{11}{52}\right) + 3\left(-\frac{11}{52}\right) = 5.2 + \left(-\frac{11}{52}\right)$$

$$\frac{41}{10} + \frac{99}{65} - \frac{33}{52} = \frac{1,297}{260}$$

$$\frac{1,297}{260} = \frac{1,297}{260} \checkmark$$

10. $\frac{1}{4}x - 3(x - 2) = -2.25x + 1$

$$0.25x - 3x + 6 = -2.25x + 1$$

$$-2.75x + 2.25x = 1 - 6$$

$$-0.5x = -5$$

$$x = 10$$

$$\frac{1}{4}(10) - 3(10 - 2) = -2.25(10) + 1$$

$$2.5 - 3(8) = -22.5 + 1$$

$$2.5 - 24 = -21.5$$

$$-21.5 = -21.5 \checkmark$$